Literature

Wiki *Enhanced dissipation for passive scalars*

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1 Orientation

1. M. Coti Zelati, G. Crippa, G. Iyer and A. L. Mazzucato, *Mixing in incompressible flows: transport, dissipation, and their interplay*, Notices of the American Mathematical Society **71** (2024), 593–604, arXiv:2308.00358.

2. A. L. Mazzucato, Y. Feng and C. Nobili, *Enhanced dissipation by advection and applications to PDEs*, Applied Mathematical Modelling **150** (2026), Part A, 116310, arXiv:2501.17695.

2 General mechanisms and rate limitations

1. P. Constantin, A. Kiselev, L. Ryzhik and A. Zlatoš, *Diffusion and mixing in fluid flow*, Annals of Mathematics **168** (2008), 643–674, arXiv:math/0509663.

2. W. Cooperman, G. Iyer, K. Rowan and S. Son, *Exponentially mixing flows with slow enhanced dissipation*, preprint (2025), arXiv:2507.21305.

3. M. Coti Zelati, M. G. Delgadino and T. M. Elgindi, *On the relation between enhanced dissipation timescales and mixing rates*, Communications on Pure and Applied Mathematics **73** (2020), 1205–1244, arXiv:1806.03258.

4. Y. Feng and G. Iyer, *Dissipation enhancement by mixing*, Nonlinearity **32** (2019), 1810–1851, arXiv:1806.03699.

5. C. Seis, *Bounds on the rate of enhanced dissipation*, Communications in Mathematical Physics **399** (2023), 2071–2081, arXiv:2003.08794.

6. D. Wei, *Diffusion and mixing in fluid flow via the resolvent estimate*, Science China Mathematics **64** (2021), 507–518, arXiv:1811.11904.

3 Smooth stationary shears

1. D. Albritton, R. Beekie and M. Novack, *Enhanced dissipation and Hörmander's hypoellipticity*, Journal of Functional Analysis **283** (2022), 109522, arXiv:2105.12308.

2. J. Bedrossian and M. Coti Zelati, *Enhanced dissipation, hypoellipticity, and anomalous small noise inviscid limits in shear flows*, Archive for Rational Mechanics and Analysis **224** (2017), 1161–1204, arXiv:1510.08098.

3. M. Coti Zelati, *Stable mixing estimates in the infinite Péclet number limit*, Journal of Functional Analysis **279** (2020), 108562, arXiv:1909.01310.

4. M. Coti Zelati and T. D. Drivas, *A stochastic approach to enhanced diffusion*, Annali della Scuola Normale Superiore di Pisa, Classe di Scienze **22** (2021), 811–834, arXiv:1911.09995.

5. M. Coti Zelati and T. Gallay, *Enhanced dissipation and Taylor dispersion in higher-dimensional parallel shear flows*, Journal of the London Mathematical Society **108** (2023), 1358–1392, arXiv:2108.11192.

4 Circular and Hamiltonian flows and Taylor dispersion

1. M. Beck, O. Chaudhary and C. E. Wayne, *Analysis of enhanced diffusion in Taylor dispersion via a model problem*, Hamiltonian Partial Differential Equations and Applications, Fields Institute Communications **75** (2015), pp. 31–71, arXiv:1501.05877.

2. M. Beck, O. Chaudhary and C. E. Wayne, *Rigorous justification of Taylor dispersion via center manifolds and hypocoercivity*, Archive for Rational Mechanics and Analysis **235** (2020), 1105–1149, arXiv:1804.06916.

3. E. Bruè, M. Coti Zelati and E. Marconi, *Enhanced dissipation for two-dimensional Hamiltonian flows*, Archive for Rational Mechanics and Analysis **248** (2024), Article 84, arXiv:2211.14057.

4. M. Coti Zelati and M. Dolce, *Separation of time-scales in drift-diffusion equations on R^2*, Journal de Mathématiques Pures et Appliquées **142** (2020), 58–75, arXiv:1907.04012.

5. M. Coti Zelati, M. Dolce and C. Lo, *Diffusion enhancement and Taylor dispersion for rotationally symmetric flows in discs and pipes*, Journal of Mathematical Fluid Mechanics **26** (2024), Article 12, arXiv:2305.18162.

6. M. Dolce, C. J. P. Johansson and M. Sorella, *Dissipation enhancing properties for a class of Hamiltonian flows with closed streamlines*, Communications in Partial Differential Equations **50** (2025), 301–352, arXiv:2407.06884.

7. Y. Feng, A. L. Mazzucato and C. Nobili, *Enhanced dissipation by circularly symmetric and parallel pipe flows*, Physica D: Nonlinear Phenomena **445** (2023), 133640, doi:10.1016/j.physd.2022.133640.

8. G. Iyer and H. Zhou, *Quantifying the dissipation enhancement of cellular flows*, SIAM Journal on Mathematical Analysis **55** (2023), 6496–6516, arXiv:2209.11645.

9. T. Li and L. Zhang, *Enhanced dissipation and Taylor dispersion by a parallel shear flow in an infinite cylinder with unbounded cross section*, preprint (2025), arXiv:2510.13097.

5 Time dependent shears and localized estimates

1. J. Bedrossian, S. He, S. Iyer and F. Wang, *Pseudo-Gevrey smoothing for the passive scalar equations near Couette*, Journal of Functional Analysis **289** (2025), 110987, arXiv:2405.19233.

2. J. Benthaus, G. M. Coclite and C. Nobili, *Mixing and enhanced dissipation in a time-translating shear flow*, preprint (2026), arXiv:2603.14624.

3. J. Benthaus and C. Nobili, *Enhanced dissipation via time-modulated velocity fields*, Evolution Equations and Control Theory **15** (2026), 21–50, arXiv:2501.16905.

4. D. Coble and S. He, *A note on enhanced dissipation of time-dependent shear flows*, Communications in Mathematical Sciences **22** (2024), 1685–1700, arXiv:2309.15738.

5. S. He, *Localized Enhanced Dissipation: A Hypocoercivity Approach*, preprint (2026), arXiv:2603.14657.

6 Deterministic mixing constructions

1. T. M. Elgindi, K. Liss and J. C. Mattingly, *Optimal enhanced dissipation and mixing for a time-periodic, Lipschitz velocity field on T2*, Duke Mathematical Journal **174** (2025), 1209–1260, arXiv:2304.05374.

7 Rough velocities and generalized diffusion

1. M. Colombo, M. Coti Zelati and K. Widmayer, *Mixing and diffusion for rough shear flows*, Ars Inveniendi Analytica, Paper No. **2** (2021), 21 pp., arXiv:2009.12268.

2. Y. Feng and A. L. Mazzucato, *Dissipation enhancement by shear flows for generalized diffusion operators in high dimension*, Communications in Mathematical Sciences **23** (2025), 2203–2215, doi:10.4310/CMS.250930031923.

3. L. Galeati and M. Gubinelli, *Mixing for generic rough shear flows*, SIAM Journal on Mathematical Analysis **55** (2023), 7240–7272, arXiv:2107.12115.

4. S. He, *Enhanced dissipation, hypoellipticity for passive scalar equations with fractional dissipation*, Journal of Functional Analysis **282** (2022), 109319, arXiv:2103.07906.

5. M. Romito and L. Roveri, *Instantaneous Total Enhanced Dissipation For Very Rough Shear Flows*, preprint (2025), arXiv:2510.27331.

8 Random stirring and transport noise

1. J. Bedrossian, A. Blumenthal and S. Punshon-Smith, *Almost-sure enhanced dissipation and uniform-in-diffusivity exponential mixing for advection-diffusion by stochastic Navier-Stokes*, Probability Theory and Related Fields **179** (2021), 777–834, arXiv:1911.01561.

2. A. Blumenthal, M. Coti Zelati and R. S. Gvalani, *Exponential mixing for random dynamical systems and an example of Pierrehumbert*, Annals of Probability **51** (2023), 1559–1601, arXiv:2204.13651.

3. W. Cooperman, G. Iyer and S. Son, *A Harris theorem for enhanced dissipation, and an example of Pierrehumbert*, Nonlinearity **38** (2025), 045027, arXiv:2403.19858.

4. M. Coti Zelati, M. Hairer and D. Villringer, *A Stochastic RAGE Theorem and Enhanced Dissipation for Transport Noise*, preprint (2025), arXiv:2507.11422.

5. A. Del Zotto and M. Nualart, *Exponential mixing and enhanced dissipation on the unit sphere with Rossby-Haurwitz flows*, preprint (2026), arXiv:2606.11055.

6. F. Flandoli and D. Luo, *Enhanced dissipation and Lyapunov exponents for stochastic transport-diffusion equations with small molecular diffusivity and small noise intensity*, Stochastics and Partial Differential Equations: Analysis and Computations **14** (2026), 1062–1076, doi:10.1007/s40072-026-00426-y.

9 Inviscid mixing anomalous dissipation and further lower bounds

1. G. Alberti, G. Crippa and A. L. Mazzucato, *Exponential self-similar mixing by incompressible flows*, Journal of the American Mathematical Society **32** (2019), 445–490, arXiv:1605.02090.

2. G. Crippa, T. Elgindi, G. Iyer and A. L. Mazzucato, *Growth of Sobolev norms and loss of regularity in transport equations*, Philosophical Transactions of the Royal Society A **380** (2022), 20210024, arXiv:2109.14975.

3. T. D. Drivas, T. M. Elgindi, G. Iyer and I. Jeong, *Anomalous dissipation in passive scalar transport*, Archive for Rational Mechanics and Analysis **243** (2022), 1151–1180, arXiv:1911.03271.

4. T. M. Elgindi and A. Zlatoš, *Universal mixers in all dimensions*, Advances in Mathematics **356** (2019), 106807, arXiv:1809.09614.

5. T. M. Elgindi and K. Liss, *Norm growth, non-uniqueness, and anomalous dissipation in passive scalars*, Archive for Rational Mechanics and Analysis **248** (2024), arXiv:2309.08576.

6. G. Iyer, A. Kiselev and X. Xu, *Lower bounds on the mix norm of passive scalars advected by incompressible enstrophy-constrained flows*, Nonlinearity **27** (2014), 973–985, arXiv:1310.2986.

7. C. Nobili and S. Pottel, *Lower bounds on mixing norms for the advection diffusion equation in Rd*, Nonlinear Differential Equations and Applications NoDEA **29** (2022), Article 12, arXiv:2006.04614.